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Thermal performance optimization of a fuel element in particle bed reactors for nuclear thermal propulsion

Yu Ji[a], Han Zhang[b], Jun Sun[a,*], Lei Shi[a]

[a] Institute of Nuclear and New Energy Technology, Collaborative Innovation Center of Advanced Nuclear Energy Technology, Key Laboratory of Advanced Reactor Engineering and Safety of Ministry of Education, Tsinghua University, Beijing 100084, China
[b] Institute of Nuclear and Energy Technologies, Karlsruhe Institute of Technology, Hermann-von-Helmholtz-Platz 1, 76344 Eggenstein-Leopoldshafen, Germany

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ABSTRACT

Nuclear Thermal Propulsion (NTP) provides the leading potential due to its high thrust, improved specific impulse and long accumulated lifetime in the manned deep space exploration. Particle Bed Reactor (PBR) is the most efficient one among all proposed NTP concepts, which employs a radial flow pattern to reduce the pressure drop and fuel particles of high surface-to-volume ratio to increase the heat removal capability. The technical challenges, however, have arisen with the enhanced performance of PBR, such as the thermal hydraulic design. In this paper, a procedure based on the Non-dominated Sorting Genetic Algorithm with elitisms approach (NSGA-2) and uncertainty quantification is developed to prepare an optimal fuel element for PBR with regard to the thermal performance. The procedure has been verified through three cases, in which the thermal performance of all fuel elements has improved a lot after the optimization process.

1. Introduction

Nowadays humans are focusing their attention on the exploration throughout the solar system and beyond, including the establishment of the colony on other planets (Nam et al., 2015). Mars may be the first one, and the manned mission to Mars has been clearly illustrated in the Human Exploration of Mars Design Reference Architecture 5.0 (DRA 5.0) by NASA in 2009 (Drake et al., 2010). To ensure a successful Martian colony, there are two major challenges that should be overcome. One is the prevention of the astronauts from damage in the harsh space environment (Nam et al., 2016), the other is the reliable supply of food, material and energy to the Mars surface, especially in the initial stages (Borowski et al., 2012). Hence it is required to develop a feasible transportation system of rapid orbital transfer, reliability, safety and reusability. Nuclear thermal propulsion (NTP) is a promising technology. It uses fission energy rather than the combustion of a fuel-oxidizer mixture to heat the liquid hydrogen propellant for rocket thrust, and has been adopted for the cargo and crew transfer from low Earth orbit (LEO) to Mars orbit as well as the return of crew to Earth in the DRA 5.0 (Drake et al., 2010), i.e., the step 2, 8 and 13 in the mission sequence for manned Mars exploration, as shown in Fig. 1. Due to the lowest molecular weight of hydrogen on the Earth, NTP has at least twice the specific impulse of the most advanced chemical propulsion systems currently (Gates et al., 2018; Labib and King, 2015). Besides, it can reach tons of thrust, which offers a competitive acceleration over other in-space electrical thrusters (Venneri and Kim, 2015). Therefore, NTP can achieve a shorter stay duration in space, almost 180 days in both outbound and inbound trips for crew transfer. In addition, coupled with the longer sustained lifetime, NTP becomes the preferred option in the alternative propulsion techniques for interstellar traveling consequently.

NTP was firstly proposed in 1940s~1950s, and has been rapidly developed in the subsequent decades with regard to the conceptual design, ground test and even flight ready systems (Gabrielli and Herdrich, 2015; Idaho, 2015; Robbins, 1991). During this period, the most representative efforts were the Nuclear Engine Rocket Vehicle Application (NERVA) in the United States of America (Clark et al., 1993) and the Nuclear Engine RD-0410 in the Union of Soviet Socialist Republics (Zakirov and Pavshook, 2007). Hexagonal elements with drilled holes were selected in the NERVA core, while twisted ribbon fuel elements and assembled bundles were adopted in the RD-0410 reactor core. The coolant flows axially and gets heated in these two reactor cores. A large fund was poured into the NERVA and RD-0410, but they were both aborted for some non-technical reasons before the launching. Fortunately, NTP has sprung to life and attracted much interest in recent years. The experience and achievements from NERVA program made its core become the foundation of many research and the prototype of some new designs by Venneri and Kim (2015), Houts and

* Corresponding author.
E-mail address: sunjun@tsinghua.edu.cn (J. Sun).

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Nomenclature

a	coefficient in the shape function of porosity profile
b	relative axial position corresponding to maximum porosity
c	relative maximum porosity on the cold frit
d	diameter, m
f	coefficient of pressure loss in pebble bed
h	heat transfer coefficient, $W\,m^{2}\,K^{-1}$
L	length of the fuel element, m
m	mass flow rate, $kg\,m^{2}\,s^{-1}$
Nu	Nusselt number, $=hd/\lambda$
Pr	Prandtl number, $=\mu c_p/\lambda$
Re	Reynolds number, $=\rho Ud/\mu$
T	temperature, K
u	velocity, $m\,s^{-1}$
x	location in coordinates, m

Greek letters

α	porosity of particle bed
β	confidence level
γ	fraction of population within tolerance limit
ε	porosity on the cold frit
λ	thermal conductivity, $W\,m^{-1}\,K^{-1}$
μ	dynamic viscosity, $kg\,m^{-1}\,s^{-1}$

Subscripts and superscripts

c	cold frit
h	hot frit
max	maximum value
out	value at outlet
p	particle

Abbreviation

CFD	computational fluid dynamics
LHS	latin hypercube sampling
MOO	multi objective optimization
NERVA	nuclear engine for rocket engine application
NSGA-2	non-dominated sorting genetic algorithm with elitism
NTP	nuclear thermal propulsion
PBR	particle bed reactor
PDF	probability distribution function
UQ	uncertainty quantitation

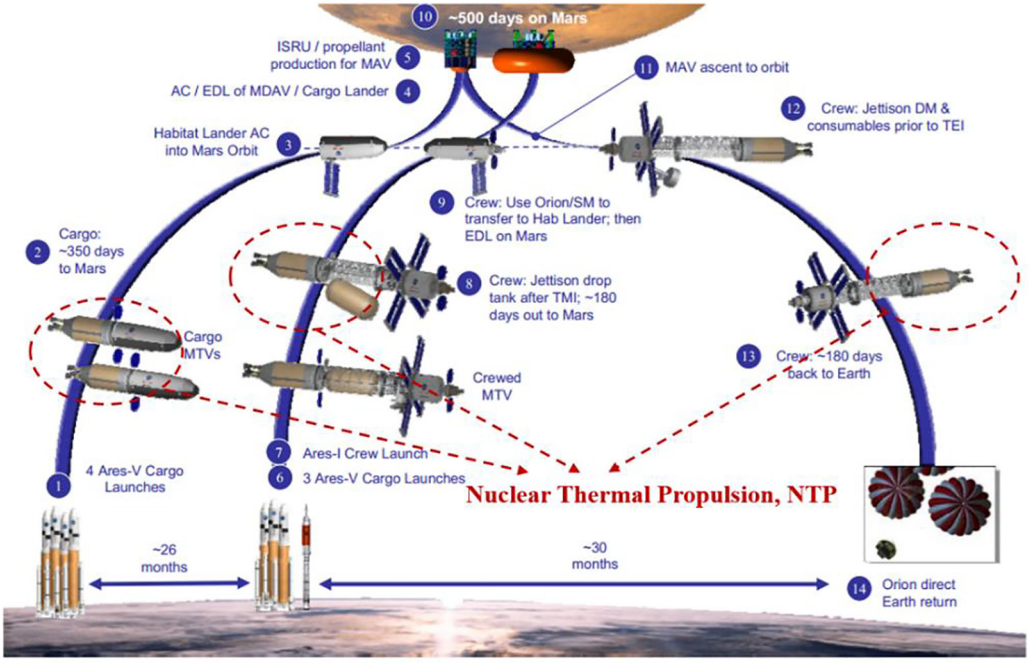


Fig. 1. Mission sequence for manned Mars exploration in DRA 5.0 (Drake et al., 2010).

Mitchell (2016), Gates et al. (2018), and so on. In addition, the use of clustered NERVA engines was also addressed in the DRA 5.0 to provide an "engine-out" capability that could reduce the mission risk and increase crew safety.

Accompanied by the new life of NERVA core, several novel concepts have also been proposed, among which particle bed reactor (PBR) is the most efficient one (Ludewig et al., 1996). Different from the NERVA core or the RD-0410 reactor core, PBR employs a radial flow configuration in the fuel element (Houts and Mitchell, 2016), as shown in Fig. 2. For each fuel element, the fuel particles (almost 500 μm in diameter) are packed in the annular region between cold frit and hot frit, through which the hydrogen flows and gets heated (Emrich, 2016). Compared to the NERVA core or the RD-0410 core, the pressure drop within the PBR core is lower due to shorter flow path of the coolant. In addition, fuel particle is of higher surface-to-volume ratio than any other fuel forms, which enhances the heat transfer capability and enables an extremely high power density in the nuclear core, potentially leading to a compact design and high thrust-to-weight ratio. Hence the capability for manned Mars exploration could be enhanced further if PBR is realized. However, the performance improvement and technical challenges coexist in the PBR. A few researches have been performed to investigate the flow and heat transfer characteristic of PBR (Morley et al., 1994; Morley and El-Genk, 1994), but no clear approach for thermal design was found in the open literature. According to the previous investigations by the authors, it was found that the flow distribution of coolant along the axial direction of the fuel element is one

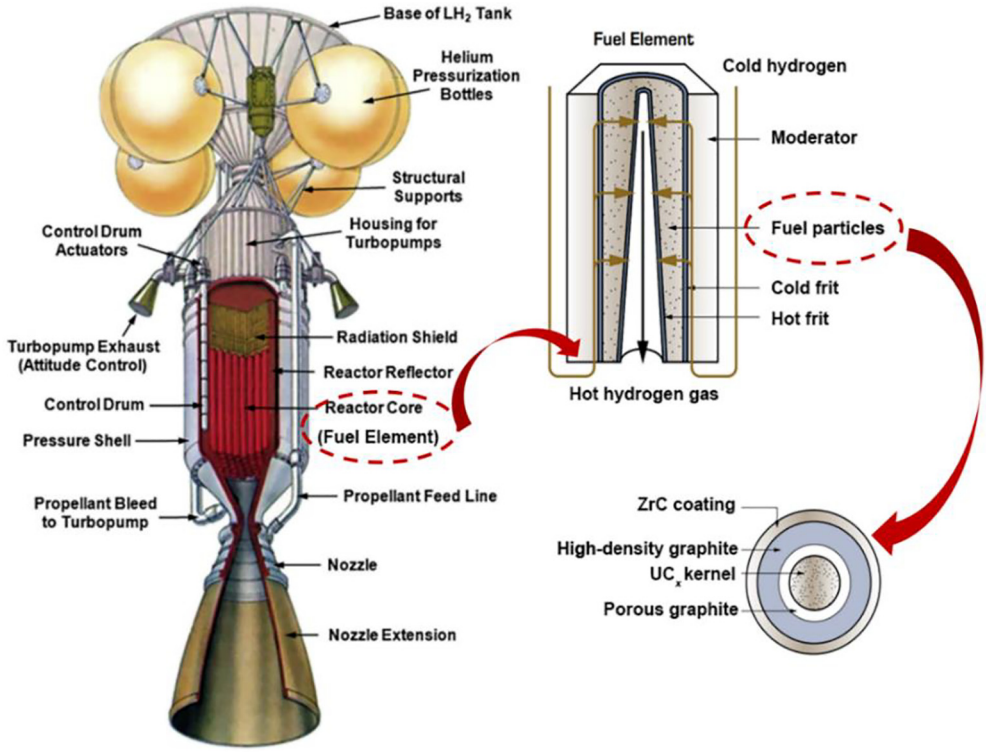


Fig. 2. Particle bed reactor based NTP system and its components (Houts and Mitchell, 2016).

of the most challenging problems in designing the PBR, and cold frit could affect the flow distribution and the consequent thermal performance of the fuel element a lot. Moreover, the authors have also proposed an analytical approach to design a fuel element of uniform power profile in the stages of preliminary design. By using this approach, the porosity on the cold frit can only be designed as a constant. However, if the power varies along the axial direction, the design of a constant porosity on the cold frit may lose its effectiveness and the porosity profile on the cold frit should be optimized to reduce both the maximum temperature and the pressure drop in the fuel element.

To get an optimal design, there are many available methods, which are generally divided into two categories, i.e., iterative methods and heuristic methods. Iterative methods include Newton's method, gradient descent method, conjugate gradient method and so on, which need the evaluation of the Hessian matrix or the gradient of the objective function in a prior and cost much effort. Heuristic methods include the genetic algorithms (GA) (Pinheiro and Schirru, 2019), particle swarm optimization (PSO) (Lim et al., 2019) and so on, which only need objective function values themselves. Besides, the problem in the present work involves two objectives, i.e., maximum temperature and pressure drop, which is a multi-objective optimization (MOO). Iterative methods are mainly used in the optimization of only one objective problem. The multi objective can be transformed to one objective through weighting method, but the weighting technique relies on the subjective judgements greatly. However, GA can efficiently obtain the optimal configuration and have significant practical value in the MOO problems (Han et al., 2019). By using GA, a Pareto-optimal front including several designs is obtained rather than only one certain design, and you can select anyone in this front according to your considerations. The non-dominated sorting genetic algorithm with elitisms approach (NSGA-2) is a common algorithm (Deb et al., 2002), and has been widely used to find the optimal conditions in many areas due to its easy implement and reduced computational complexity compared to the traditional NSGA method (Bu et al., 2018; Han et al., 2019), thus it is selected in the present paper to optimize the porosity profile on the cold frit to improve the thermal performance of the fuel element.

Usually, the optimization is performed at a certain condition, but the actual parameters may be perturbed because of the operation variation or manufacturing error. Therefore, it is required to evaluate whether the design could tolerate the perturbations of some parameters and maintain the excellent performance. For the nuclear reactor design, the peak temperature and the pressure drop should not exceed the constrains at any conditions to ensure the safety (Avramova et al., 2015; Avramova and Ivanov, 2010). Uncertainty quantification (UQ) based on the random sampling or deterministic sampling is a frequently-used method to estimate the tolerance limit under the parameters' perturbation (Zhang et al., 2018). To date, it has been applied to analyze the performance of a Pebble Bed HTGR in the loss of cooling event (Strydom, 2013), evaluate the safety of a Sodium-Cooled Fast Reactor in some unprotected accidents (Morris and Nutt, 2011), and investigate the contribution of several parameters to the peaking fuel temperature of a Fluoride-salt-cooled High-temperature Reactor in the steady state and unprotected loss of flow transient (Wang et al., 2017), etc. Thus UQ shows a great potential to evaluate the performance of the derived optimal design of fuel element under some parameters' perturbation.

This paper aims to develop a procedure to prepare an optimal fuel element for PBR. By using NSGA-2 approach, the porosity profile on the cold frit is optimized to match the power distribution and improve the thermal performance in the fuel element. After getting the optimal design, the performance is re-evaluated under perturbed conditions through UQ.

2. Description and simplification of the fuel element

The fuel element of PBR have been introduced briefly in the previous section. Some supplemental information is reported here. The hexagonal fuel element consists of moderator base, inlet plenum, cold frit, fuel particle bed, hot frit and hot gas channel along the inward radial direction (Ludewig et al., 1996). The moderator base is usually manufactured by zirconium hydride (ZrH) or lithium hydride (LiH) and provides neutron moderation. The cold frit and hot frit are two

concentric orifices made by light metal and refractory metal respectively, and can be used to adjust the flow distribution (Emrich, 2016). Generally, it is better to use cold frit to distribute the coolant in the fuel element of high power density, i.e., a cold frit dominated design, while use hot frit to distribute the coolant in the fuel element of low power density, i.e., a hot frit dominated design. Many coated fuel particles are packed between the two frits. The coolant enters the fuel element through inlet plenum and leaves from the hot gas channel by two flow configurations. One is "U" flow pattern, which sets the inlet and outlet at the same side of the fuel element; the other is "Z" flow pattern, which sets the inlet and outlet at the different sides of the fuel element. In this paper, the fuel element without moderator base is simplified as a 2D model in cylindrical coordinates due to the symmetry, and "Z" flow pattern is selected for the later convenient connection to thrust chamber. The schematic view of the simplified fuel element is shown in Fig. 3.

3. Methodology and procedure

3.1. Optimization procedure

According to the previous investigation by the authors, the thermal performance of the fuel element is significantly affected by its structure and the porosity profile on the frits (Ji et al., 2018; Ji et al., 2019). Hence a stepwise procedure including preliminary design, optimization and evaluation is developed to prepare an optimal fuel element for the PBR, as shown in Fig. 4. In the preliminary design phase, the power in the fuel region and the porosity on both the cold and hot frits are all assumed as a uniform profile, and the objective is to determine the major structural parameters such as the length of the fuel element, thickness of the particle bed, inner radius of the hot gas channel. Once this preliminary design is obtained, the following steps can be facilitated to get an optimal fuel element under non-uniform power profile.

1. Establish a shape function of the porosity profile on cold frit, of which the coefficients should be determined by the MOO.
2. Optimize the shape function to get a fuel element of excellent thermal hydraulic performance under the non-uniform power profile by using NSGA-2.
3. Evaluate the tolerance limit of important parameters of this derived optimal fuel element under some perturbations through UQ.

3.2. Shape function of porosity profile on cold frit

As introduced previously, the cold frit affects the thermal hydraulic performance of the fuel element a lot and an optimal design contributes to the reduction of peak temperature and pressure drop in the fuel element. Here a general shape function is established to specify the variation of the porosity profile along the axis, and expressed as:

$$\varepsilon = \varepsilon_0 + \varepsilon_1 + \varepsilon_2 \tag{1}$$

in which ε_0 is the base porosity; ε_1 is the porosity accounting for the contribution of the top reflector, while ε_2 is the porosity accounting for the contribution of the bottom reflector. For a fuel element without top and bottom reflector, the heat release always presents as a convex profile, which is of peak value in the middle while of lower value at the end of the fuel element due to the neutron leakage. The simplest equation matching this profile is the parabolic function, thus it is adopted to describe the variation of base porosity, and written as:

$$\varepsilon_0 = (a(x - b{\cdot}L)^2 + c){\cdot}\alpha_0 \tag{2}$$

in which b corresponds to the relative axial position of maximum porosity, c is the relative maximum porosity, L is the length of the fuel element, α_0 is the mean porosity, x is the real local axial position. By assuming the mean relative porosity equals to 1, i.e.,

$$\frac{1}{L}\int_0^L [a(x - b{\cdot}L)^2 + c]\, dx = 1 \tag{3}$$

the coefficient a can be determined as:

$$a = \frac{3(1-c)}{[b^3 + (1-b)^3]{\cdot}L^2} \tag{4}$$

For a fuel element with top or bottom reflector, the power density increases significantly in the region adjacent to the reflectors due to the additional neutron moderation, and decreases rapidly when away from the reflectors. Therefore, the exponential functions are selected to model the porosity profile accounting for the contribution of top and bottom reflectors, which yield as:

$$\varepsilon_1 = \alpha_1 e^{-\lambda_1 \frac{x}{L}} \tag{5}$$

and

$$\varepsilon_2 = \alpha_2 e^{-\lambda_2\left(1-\frac{x}{L}\right)} \tag{6}$$

respectively, where α_1 and α_2 represent the additional porosity at $x = 0$ and $x = L$ separately; λ_1 and λ_2 are the parameters characterizing the decrease rate of the additional porosity.

3.3. Optimization model of the fuel element for PBR

This paper establishes the fuel element optimization model as shown in the following equations:

$$\begin{aligned} &\min \ f = \{T_{\max},\ \Delta p\} \\ s.\,t.\quad &T_{\max} \leqslant T_{\text{limit}},\ \Delta p \leqslant \Delta p_{\text{limit}} \\ &x_i^1 \leqslant x_i \leqslant x_i^2 \end{aligned} \tag{7}$$

where $T_{\max}$ and Δp are the maximum temperature and the pressure drop in the fuel element separately. $x = \{b, c, \alpha_0, \alpha_1, \lambda_1, \alpha_2, \lambda_2\}$ in the shape function of the porosity on the cold frit, and x_i^1, x_i^2 are the lower and upper limit values of the design parameters respectively.

In the Eq. (7), the values of objective function, i.e., maximum temperature and the pressure drop in the fuel element are determined by the computational fluid dynamics (CFD) code, ANSYS FLUENT v19.2. Here the coolant flow is treated as incompressible turbulence, in which the quantities can be determined by solving the Reynolds Averaged Navier-Stokes equation and realizable k-ε transport equations. Besides, the cold frit and hot frit are both regarded as porous media, and the pressure drop within these two frits are estimated by the well-known Ergun model (Ergun, 1952):

$$\frac{\Delta p}{\Delta r} = \frac{150\mu}{d_p^2}\frac{(1-\varepsilon)^2}{\varepsilon^3}u + \frac{1.75\rho}{d_p}\frac{(1-\varepsilon)}{\varepsilon^3}u\,|u| \tag{8}$$

where d_p is the pore size, ε is the porosity and u is the superficial velocity flowing through the media. The particle bed is treated as porous media as well, where the pressure drop is determined, however, by the model in Safety Standard KTA 3102 "Reactor Core Design for High-Temperature Gas-Cooled Reactors", which reads as (Geschäftsstelle des Kerntechnischen Ausschusses beim Bundesamt für Strahlenschutz, 1981):

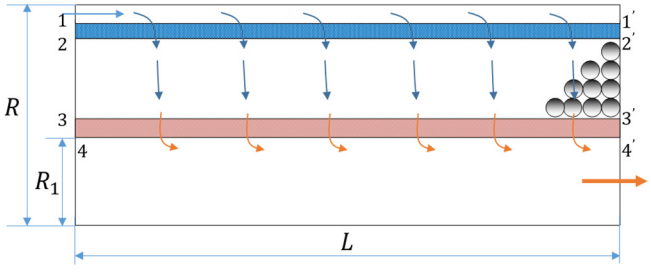


Fig. 3. Schematic view of the simplified fuel element.

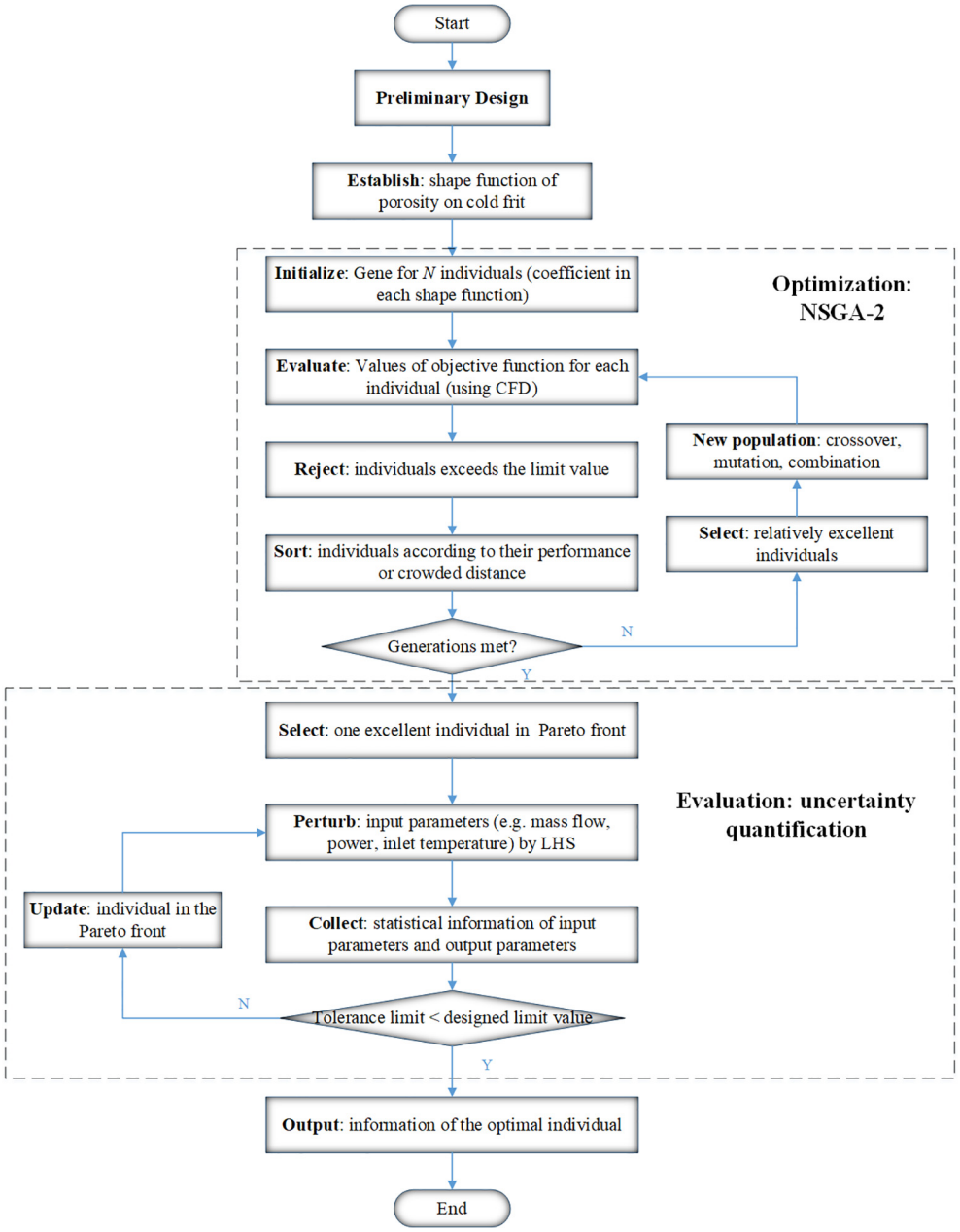


Fig. 4. Optimization procedure of fuel element for PBR.

$$\frac{\Delta p}{\Delta r} = f\frac{1-\alpha}{\alpha^3}\frac{1}{d_p}\frac{1}{2\rho}\left(\frac{\dot{m}}{A}\right)^2 \tag{9}$$

in which d_p is the diameter of the fuel particle, α is the porosity of the particle bed equaling to 0.38, $\dot{m}$ is the mass flow rate, A is the flow area, and f is the coefficient for evaluating pressure loss:

$$f = \frac{320}{\left(\frac{Re}{1-\alpha}\right)} + \frac{6}{\left(\frac{Re}{1-\alpha}\right)^{0.1}} \tag{10}$$

As for the heat transfer in the particle bed, the non-equilibrium thermal model is adopted to solve the temperature of coolant and the fuel particle. The heat transfer between the fluid and the fuel is also determined by another model in KTA 3102, which is written as (Geschäftsstelle des Kerntechnischen Ausschusses beim Bundesamt für Strahlenschutz, 1983):

$$Nu = 1.27\frac{Pr^{1/3}}{\alpha^{1.18}}Re^{0.36} + 0.033\frac{Pr^{1/2}}{\alpha^{1.07}}Re^{0.86} \tag{11}$$

Strictly speaking, the power distribution should be determined by solving the transport or diffusion equations of the neutrons, and it is also affected by the temperature profile in the fuel element. However, in the present work, no nuclear feedback has been taken into consideration, and only several different power distributions are adopted to verify the developed approach. The nuclear feedback and the coupling analysis of thermal hydraulics and neutronics will be performed in the further work.

All these models and the properties variation of hydrogen (Ji et al., 2018) are embedded into the solver through user defined functions (UDF), and the maximum temperature and the pressure drop are collected from the converged results from CFD code. Therefore, one individual with $x = \{b, c, \alpha_0, \alpha_1, \lambda_1, \alpha_2, \lambda_2\}$ input and $f = \{T_{max}, \Delta p\}$ output is realized.

In the NSGA-2 procedure, the population with size of N, namely parent population P_n should be generated first. Subsequently, another offspring population Q_n with size of N is also generated through the crossover and mutation to the parent population. Hence a gene pool is

derived by combining these two populations. The fitness function, i.e., maximum temperature and pressure drop in the fuel element, of each individual in this gene pool can be determined by performing the CFD calculation. Afterwards, a pre-judgement should be carried out, in which the individual whose fitness exceeds the limit value should be rejected to ensure a faster convergence rate. If the size of the gene pool is reduced to less than $N + 1$ after the rejection, new individuals should be supplemented to the pool. By employing non-dominated sorting function, the individuals in the updated pool are put into different fronts, namely F_1, F_2, F_3… according to their ranks. Coupled with the crowding distance function, which is used to avoid the accumulation of the individuals in a limited distance, a new parent population P_{n+1} with size of N is generated. The procedure and the schematic view are shown in Fig. 4 and Fig. 5, and the detailed introduction and the implementation of the algorithm can be found in the Reference (Deb et al., 2002). After several generations' evolution, the Pareto-optimal front solution of this MOO issue can be obtained.

3.4. Uncertainty qualification process

Uncertainty qualification to be performed in the present procedure is based on the uncertainty propagation of the input parameters, and the main steps are:

1. Choose the input uncertain parameters and confirm the probability density distribution (PDF) of them based on the existing literatures or previous engineering experience.
2. Generate a random sample of size M for the input parameters according to their PDF by using Latin Hypercube Sampling (LHS) methodology.
3. Perform corresponding CFD simulations using the generated input parameters, and get the expected output parameters.
4. Calculate the tolerance limit of the output parameters within a given confidence level using a parametric approach or a non-parametric approach. The parametric approach relies on the known probability of the output parameters, while the non-parametric approach only needs the sequence of the output variables in the increasing order. Due to the easy implementation and high accuracy, the non-parametric approach is adopted in the present procedure.

To ensure the fidelity of the UQ, the size of the sample should be determined carefully. For an issue of p ($p > 1$) output variables, the required minimum number M of the calculation runs to obtain one-sided tolerance limit at a given (β, γ) level can be determined by (Guba et al., 2003):

Table 1
The design parameters of the fuel element for PBR.

Parameters, unit	Value
Inlet temperature T_{in}, K	200
Expected outlet temperature T_{out}, K	3000
Expected chamber pressure p_{cham}, MPa	7.0
Mean power density Q, MW/l	20.0
Mass flow rate m (kg/s)	0.505
Temperature limit T_{limit}, K	3800.0
Pressure drop limit Δp_{limit}, kPa	600.0

$$\beta = \sum_{j=0}^{M-p} \binom{M}{j} \gamma^j (1-\gamma)^{M-j} \tag{12}$$

while M to obtain the two-sided tolerance limit can be determined by (Guba et al., 2003):

$$\beta = \sum_{j=0}^{M-2p} \binom{M}{j} \gamma^j (1-\gamma)^{M-j} \tag{13}$$

in which γ specifies the fraction of the population within the tolerance limit, and β is its confidence level. The Eq. (12) and Eq. (13) should be solved recursively to get M, e.g., $M = 207$ for a two-sided tolerance limit problem with three variables at (0.95,0.95) level.

4. Results and discussions

4.1. Preliminary design and its thermal performance

Before the optimization, a preliminary design of the fuel element has been obtained, of which the design parameters and the main geometry parameters are listed in the Table 1 and Table 2 respectively. In the Table 1, the inlet temperature is assumed to be 200 K, which is a common value of hydrogen after the preheating by the regenerative nozzle and the radial reflector of the PBR in the NTP systems. The outlet temperature is expected to be 3000 K, which corresponds to a high specific impulse of almost 950 s in the vacuum environment. The mass flow rate is 0.505 kg/s for a fuel element, thus the total mass flow rate for a reactor of 19 fuel elements is almost 10 kg/s, which can achieve a thrust of 100 kN. The mean power density should be 20.0 MW/l in the fuel region. Besides, the maximum temperature in the fuel element should not exceed the temperature limit, i.e. 3800 K, which is the melting point of the most advanced ternary carbide fuel (U, Zr, Nb) C (Nam et al., 2015). Meanwhile, the pressure drop in the fuel element should be less than 600 kPa, which is about one third of the pressure

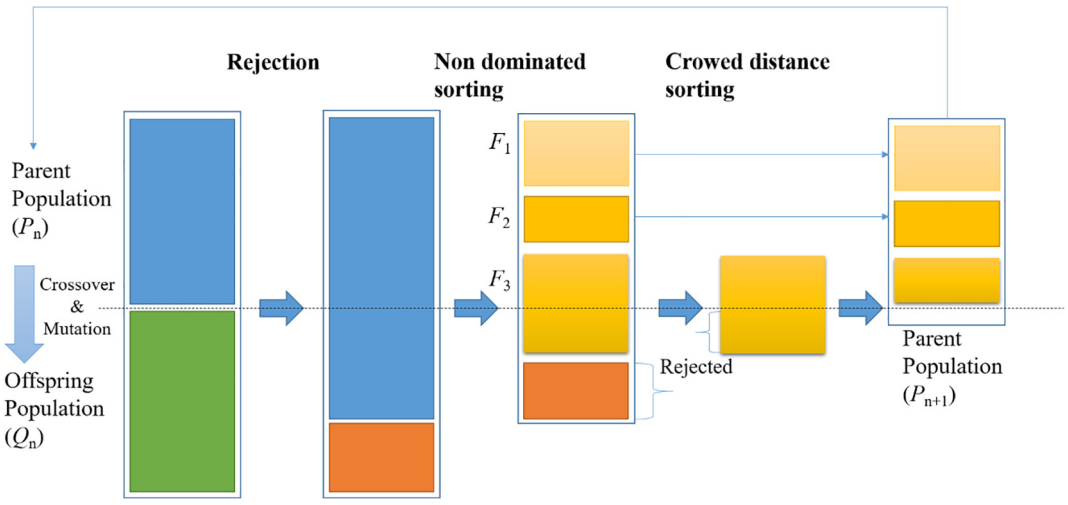


Fig. 5. Schematic view of NSGA-2 implementation for the present MOO issue.

Table 2
Main geometrical parameters of the fuel element in preliminary PBR.

Parameters, unit	Value
Radius of hot gas channel R_1, mm	25
Radius of outer hot frit R_2, mm	27
Radius of outer particle bed R_3, mm	37
Radius of outer cold frit R_4, mm	39
Radius of fuel element R_5, mm	41
Length of the fuel element L, m	0.65
Diameter of fuel particle d_p, μm	500
Porosity of fuel bed α, –	0.38
Pore size on cold frit d_c, μm	2
Porosity on cold frit ε_c, –	0.5
Pore size on hot frit d_h, μm	100
Porosity on hot frit ε_h, –	0.5

drop in the NERVA core. In the NTP systems, the greater value of the pressure drop through the system is, the more difficult we design and fabricate the hydrogen pump. By establishing the computational model using the geometrical parameters in the Table 2 and performing the CFD calculation, the maximum temperature and the pressure drop in this fuel element under a uniform power profile are derived, viz., 3348.4 K and 529.8 kPa respectively, which almost meet the design requirements shown in Table 1.

4.2. Determining the optimal design

The preliminary design is eligible in terms of its thermal performance under a uniform power distribution. However, when the power distribution changes to a non-uniform one, the peak temperature is likely to exceed the temperature limit in the fuel element. Thus the fuel element should be optimized to reach a lower peak temperature but without any significant increase of the pressure drop. In this section, three cases, including a bare fuel element of symmetric power distribution, a bare fuel element of asymmetric power distribution and a fuel element with top and bottom reflector are performed to verify the developed optimization approach.

4.2.1. A bare fuel element

4.2.1.1. Symmetric power distribution. In a bare fuel element (no reflector at the top or bottom of the fuel element), the power density can sometimes vary as a cosine function along the axial direction, which is read as:

$$F = F_m \cdot \cos\left(\frac{\pi(x - L/2)}{L_e}\right) \tag{14}$$

where F_m is the maximum value of the peak power density factor in the fuel element and is assumed to 1.25; L_e is the extrapolation length and is assumed to 1.4 L; x is the local axial position. Under this power distribution, the peak temperature in the fuel element has risen to 3905.5 K, which is significantly greater than the temperature limit. Therefore, it is required to adopt a non-uniform porosity profile instead on the cold frit to improve the thermal performance of the fuel element, and the coefficient in the shape function, i.e., Eqs. (1)–(6), should be determined by the NSGA-2 process. The lower limit value and the upper limit value of these coefficient are shown in Table 3, in which b and c are based on the power profile in the fuel element, while the parameter α_0 is based on the preliminary design and the mechanical requirements, the additional porosity α_1 and α_2 are both 0. The parameters λ_1 and λ_2 representing the decrease of the additional porosity at the top and bottom respectively are random numbers here.

In the first step of the NSGA-2 process, a parent population with size of 8 is generated randomly. In addition, the crossover and mutation probability are set to 0.9 and 0.5 separately to generate an offspring population. During the evolution, it is required almost 10 generations to get a converged result, and the convergence history is shown in Fig. 6. From the figure, it is noted that the design points are located in the bottom-left corner gradually as generation proceeds. Moreover, the Pareto-optimal front is also detected from these derived data. On this front, the maximum temperature T_{max} is of lowest value for each pressure drop Δp. It is also clearly noted that the T_{max} and Δp are always in a conflicting relationship, which is a typical characteristic of the MOO problems. Therefore, it is impossible to obtain a design of lowest peak temperature and lowest pressure drop simultaneously, and the selection of design points should be determined according the actual requirements. In this paper, the margin to the allowed maximum temperature is less than the margin to allowed pressure drop, thus the solution point A of $T_{max} = 3147.6$ K and $\Delta p = 414.4$ kPa in the Pareto-optimal front is chosen as the optimal solution.

The design parameters of the solution A are $\{b, c, \alpha_0, \alpha_1, \lambda_1, \alpha_2, \lambda_2\} = \{0.434, 1.070, 0.533, 0, -, 0, -\}$. Fig. 7 shows its relative porosity (the ratio between the total porosity and mean base porosity) profile as well as the distribution of the power peak factor in the fuel element along the axis. From the figure, it is noticed that the position of maximum porosity does not correspond to the position of peak power density, and the porosity near the outlet is less than that near the inlet, which can be explained by the significant pressure drop in the hot gas channel and the consequent flow shift phenomenon.

The comparison of the temperature distribution in the preliminary design and the optimal design under this non-uniform power distribution is shown in Fig. 8, from which it is found that the temperature distributes more uniformly after the optimization, and the peak temperature has been reduced a lot. In addition, it is observed that the temperature gradient on the hot frit is reduced as well, which decreases the local thermal stress and the possible damage.

4.2.1.2. Asymmetric power distribution. In a bare fuel element, the power density can also sometimes present as an asymmetric profile, thus a polynomial is assumed for the power peak factor, which is written as:

$$F = a_0 + a_1\left(\frac{x}{L}\right) + a_2\left(\frac{x}{L}\right)^2 + a_3\left(\frac{x}{L}\right)^3 \tag{15}$$

where a_0, a_1, a_2, and a_3 are 0.398, 1.9752, 0.5265, and −2.244, respectively. Those coefficients are determined by fitting the axial inhomogeneous factor of neutron flux density in Reference (Jiang et al., 2015) as Eq. (15). Under this power distribution, the maximum temperature in the preliminary fuel element is 3884.1 K, while the pressure drop through the fuel element is 530.6 kPa. Apparently, the maximum temperature exceeds the limit value as well and the design should be optimized. In this optimization procedure, the upper limit value and the lower limit value of the design parameters in the shape function are almost the same as those shown in Table 3. However, the upper limit value and lower limit value of b changes to 0.7 and 0.4 respectively because the peak power is located toward the bottom of the fuel element, as shown in Fig. 10. Also, the parameters related to the NSGA-2 process remain unchanged as those in the previous section.

Table 3
Limit values of the parameters in the shape function on the cold frit.

Parameters	Lower limit value	Upper limit value
Relative axial location b corresponding to maximum ε	0.3	0.5
Relative maximum porosity c	1.0	1.2
Mean base porosity α_0	0.4	0.6
Additional porosity α_1 at the top	0	0
Parameters λ_1 representing the decrease of ε_1	–	–
Additional porosity α_2 at the bottom	0	0
Parameters λ_2 representing the decrease of ε_2	–	–

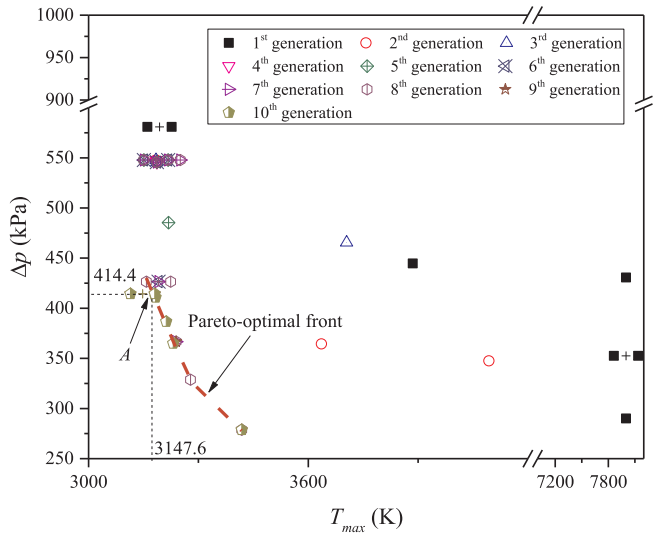


Fig. 6. Convergence history and the distribution of the Pareto-optimal solutions; a bare fuel element of symmetric power distribution.

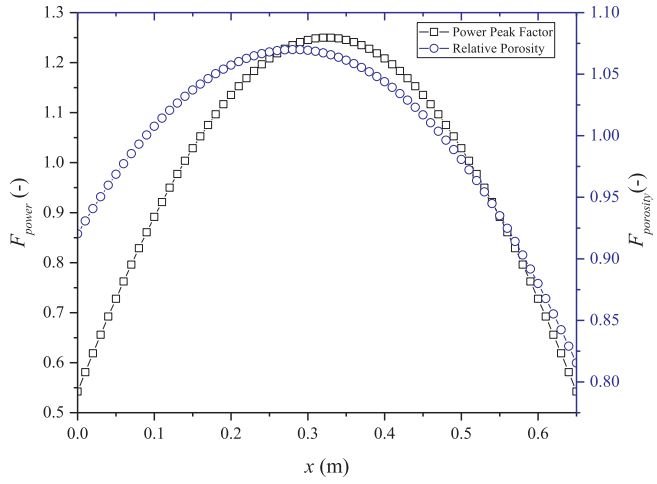


Fig. 7. Profile of the power peak factor and the relative porosity on the cold frit in the bare fuel element of symmetric power distribution.

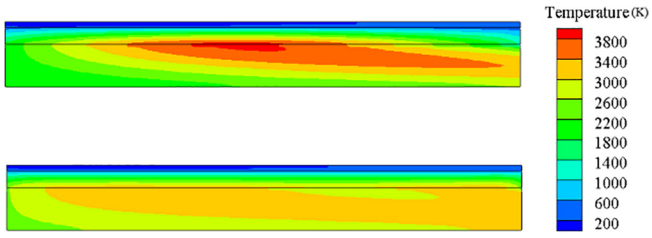


Fig. 8. Distribution of the temperature in the bare fuel element of symmetric power distribution. top, preliminary design; bottom, optimal design.

Fig. 9 shows the convergence history of the optimization procedure, from which it is noted that the solution converges and the Pareto-optimal front is derived after 12 generations' evolution. Similarly, the margin to the allowed maximum temperature is less than the margin to allowed pressure drop, thus the solution point B of $T_{max} = 3151.8$ K and $\Delta p = 312$ kPa in the Pareto-optimal front is chosen as the optimal solution.

The design parameters of the solution B are $\{b, c, \alpha_0, \alpha_1, \lambda_1, \alpha_2, \lambda_2\} = \{0.444, 1.052, 0.582, 0, -, 0, -\}$. Fig. 10 also shows the optimal relative porosity profile. Using this porosity profile, the temperature

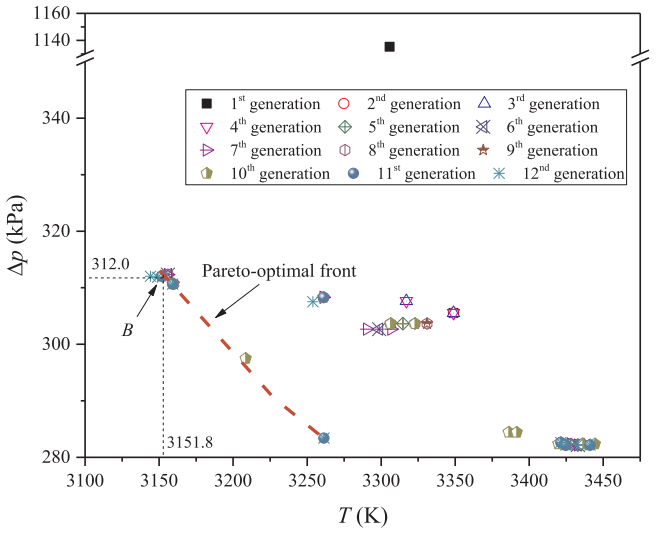


Fig. 9. Convergence history and the distribution of the Pareto-optimal solutions; a bare fuel element of asymmetric power distribution.

field in the fuel element is more uniform and flat, as shown in the Fig. 11.

4.2.2. A fuel element with top and bottom reflector

In the previous section, the effect of the optimization approach to improve the thermal performance has been discussed based on a bare fuel element of both symmetric and asymmetric power distribution. It is found that the optimization approach is effective. In this section, the performance of the optimization approach will be discussed further based on a fuel element with top and bottom reflector, where the power density increases significantly in the region adjacent to the top and bottom reflector. A polynomial is selected to fit the energy deposition derived from the neutronics design by the reactor Monte Carlo code RMC (Wang et al., 2015), which is written as:

$$F = \sum_{i=0}^{6} a_i \left(\frac{x}{L}\right)^i \tag{16}$$

where a_0, a_1, a_2, a_3, a_4, a_5, and a_6 are 1.094, −4.002, 31.445, −101.605, 166.788, −138.205, and 45.517 respectively. The variation of this power peak factor along the axis is shown in Fig. 13. Under this power distribution, the maximum temperature and the pressure drop are 3830.8 K and 529.4 kPa. Again, the maximum temperature exceeds

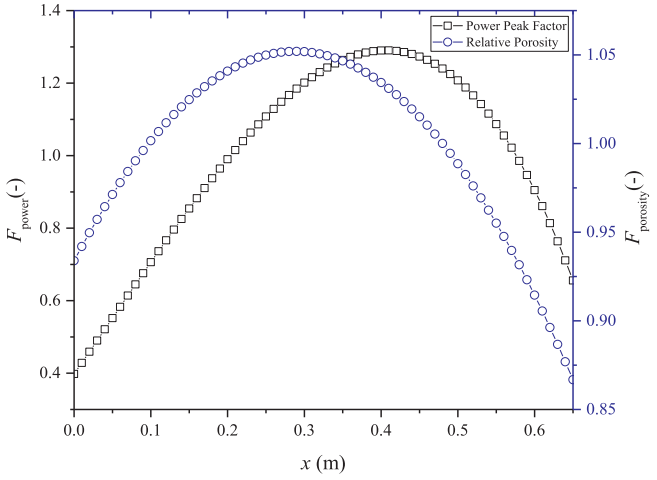


Fig. 10. Profile of the power peak factor and the relative porosity on the cold frit in the bare fuel element of asymmetric power distribution.

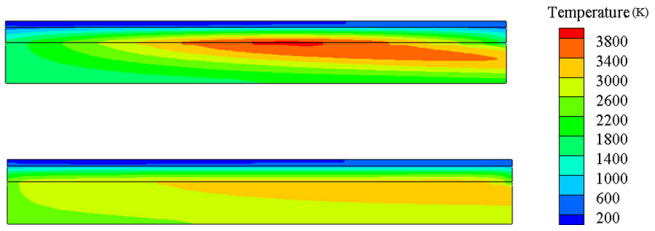


Fig. 11. Distribution of the temperature in the bare fuel element of asymmetric power distribution. top, preliminary design; bottom, optimal design.

Table 4
Limit values of the parameters in the shape function of porosity on the cold frit.

Parameters	Lower limit value	Upper limit value
Relative axial location b corresponding to maximum ε	0.4	0.5
Relative maximum porosity c	1.0	1.2
Mean base porosity α_0	0.4	0.6
Additional porosity α_1 at the top	0.1	0.4
Parameters λ_1 representing the decrease of ε_1	10	50
Additional porosity α_2 at the bottom	0.1	0.4
Parameters λ_2 representing the decrease of ε_2	10	50

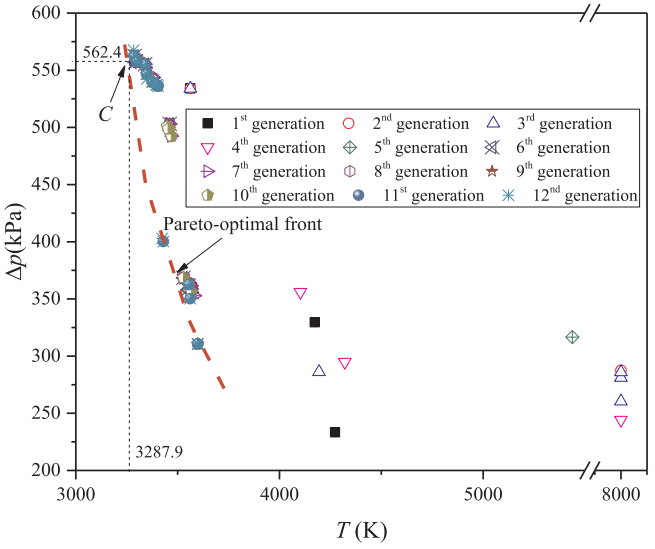


Fig. 12. Convergence history and the distribution of the Pareto-optimal solutions; a fuel element with top and bottom reflector.

the limit value and the design should be optimized as well.

In the optimization process, the lower and upper limit value of the determining coefficient are presented in Table 4, where the parameters accounting for the contribution of top and bottom reflector are determined based on the power profile. In the NSGA-2 process, the size of the parent population is raised to 16 due to the increased complexity of the power distribution. The remained coefficients in the NSGA-2 process are the same as those in the previous section. Fig. 12 shows the convergence history of this optimization procedure. From the figure, it is noted that the solution converges to the Pareto-optimal front after 12 generations' evolution. On this front, the point C of $T_{max} = 3287.9$ K and $\Delta p = 562.4$ kPa is selected as the optimal solution.

The design parameters of the solution C are $\{b, c, \alpha_0, \alpha_1, \lambda_1, \alpha_2, \lambda_2\} = \{0.418, 1.036, 0.478, 0.164, 28.470, 0.162, 32.830\}$. Fig. 13 also shows the optimal relative porosity profile. It is clearly noted that the porosity increases significantly when approaches to the top and bottom reflector. Using this porosity profile, the hot spot region has been cleared, as shown in the Fig. 14. This improved thermal performance of

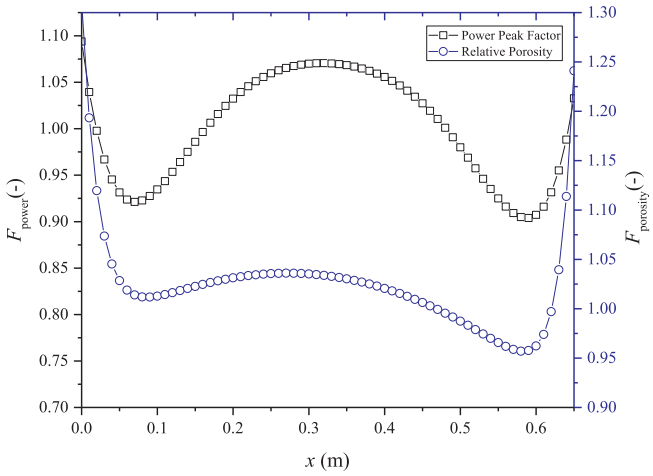


Fig. 13. Profile of the power peak factor and the relative porosity on the cold frit in the fuel element with top and bottom reflector.

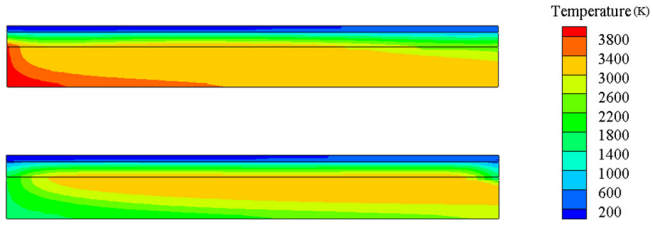


Fig. 14. Distribution of the temperature in the fuel element with top and bottom reflector. top, preliminary design; bottom, optimal design.

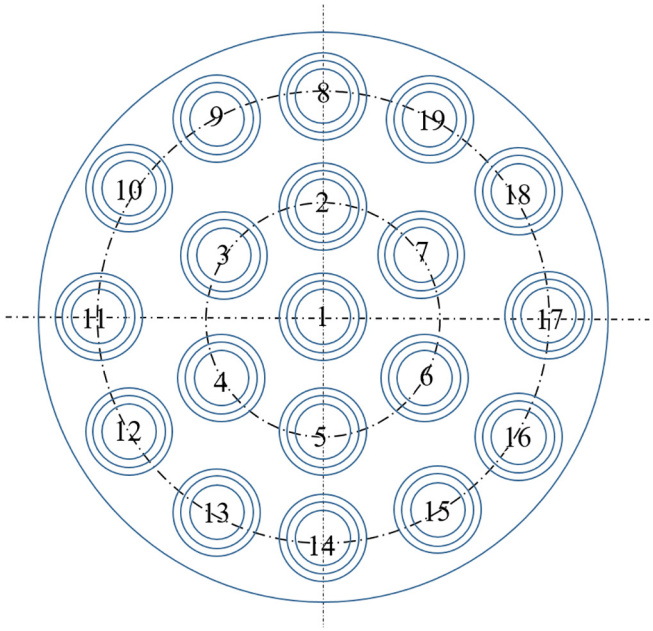


Fig. 15. Cross-sectional view of the PBR core of 19 fuel elements.

the fuel element has also verified the developed optimization approach.

4.3. Thermal performance under some parameters' perturbations

4.3.1. Input parameters selection and sampling

According to the previous experience in design the PBR, many uncertainties from the system operation, properties of the working fluid, manufacturing error and so on may affect the thermal performance of the PBR. The PBR core consists of several fuel elements, and these fuel

Table 5
Input parameters' uncertainties of the fuel element.

Index	Parameters	Mean values	2 Standard deviations (2σ)	PDF type
1	Inlet temperature	200 K	± 20 K (10%)	Normal
2	Mass flow rate	0.505 kg/s	± 0.02525 kg/s (5%)	Normal
3	Power density in the fuel element	20 MW/l	± 0.5 MW/l (2.5%)	Normal
4	Pore size on the cold frit	2.0 μm	± 0.1 μm (5%)	Normal

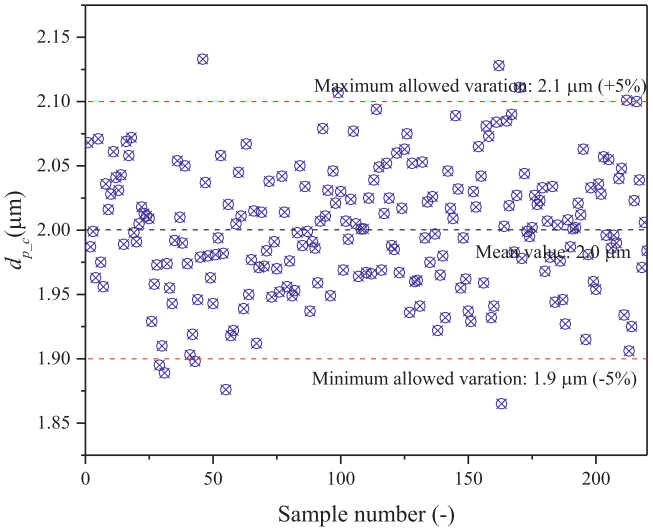


Fig. 16. Sampled pore size for 220 LHS set.

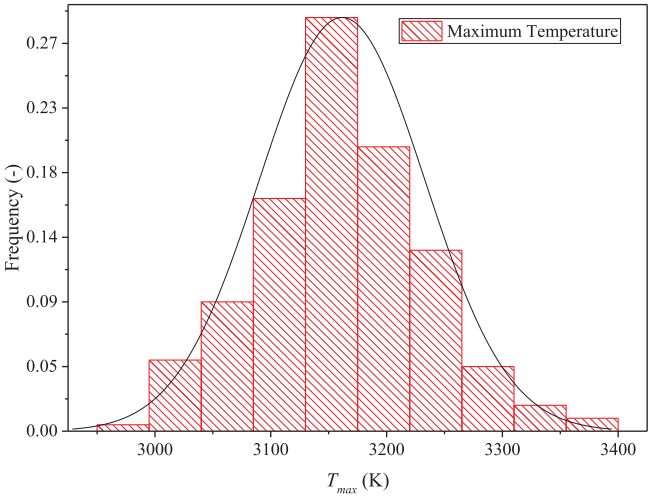


Fig. 17. Histogram of the maximum temperature.

elements can be divided into a central element and two fuel element rings for a typical core of 19 elements according to their power level, as shown in Fig. 15 (Ludewig et al., 1996). The coolant should be distributed first between these fuel elements, thus the mass flow rate may be different even for the fuel elements in the same ring. Besides, the averaged power of these elements may be different from each other due to the non-uniform profile of neutron flux along the radial and circumferential direction of the core. Moreover, the pore size on the cold frit is also important to adjust the flow distribution in the fuel element, but the uncertainty also exists in the manufacture. In this paper, therefore, four input parameters are selected for the post evaluation of the obtained optimal fuel element through UQ. They are all specified as

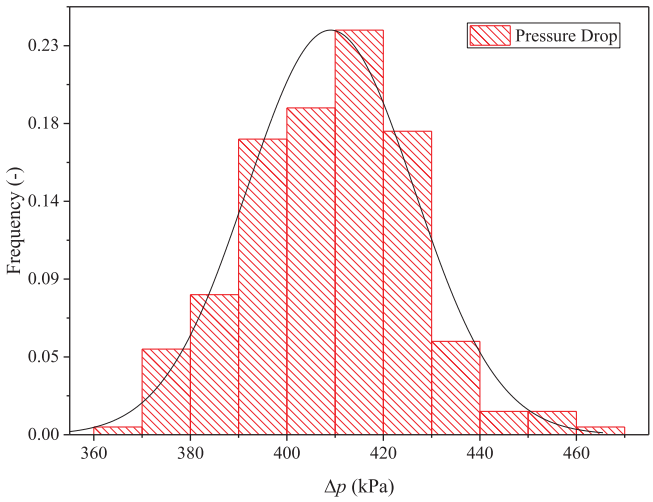


Fig. 18. Histogram of the pressure drop.

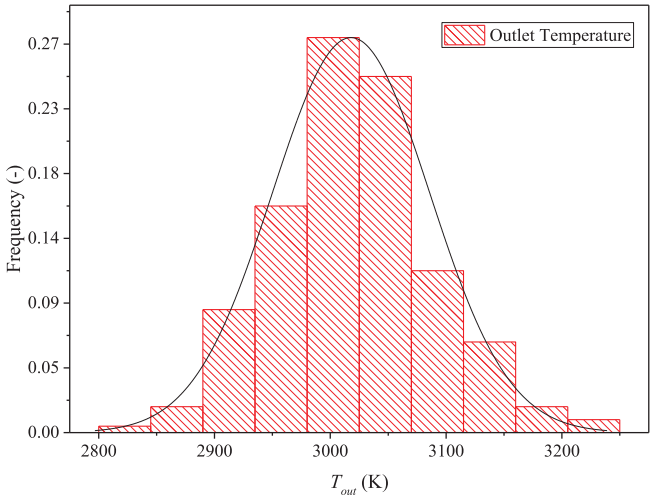


Fig. 19. Histogram of the outlet temperature.

normal distributions, of which the mean values and the standard deviation are listed in the Table 5. After the propagation of input parameters uncertainties, the two-sided tolerance limits of three variables, viz., outlet temperature, maximum temperature and the pressure drop in the fuel element are estimated, thus the required minimum size of sample is 207 according to Eq. (11). Here 220 sampling points are generated through LHS, and the scatter plot of the sampled pore size in the LHS set is showed in Fig. 16.

4.3.2. Uncertainty quantification

Take a bare fuel element of symmetric power distribution as an example and use the input uncertainties of four parameters above, the histogram of the maximum temperature and the pressure drop are shown in Fig. 17 and Fig. 18, respectively. The statistic results show that the 95% two-side tolerance limit are 2986 K and 3369 K for the maximum temperature in the fuel element with a confidence level of 95%, while 368.4 kPa and 464.3 kPa for the pressure drop in the fuel element at a same confidence level, according to the order statistic proposed by Guba et al. (2003). The upper limits of the maximum temperature and the pressure drop do not exceed the limit value even at a perturbed condition, which indicates the optimization is still effective.

In addition to the maximum temperature and the pressure drop, the histogram of the outlet temperature is also shown in Fig. 19, and the statistical information reveals that the tolerance limits are 2841 K and 3171 K at a (95%, 95%) level. The lower limit is significantly lower

than the designed outlet temperature of the fuel element, which will reduce the specific impulse of the NTP. However, this effect is related to the design of the whole PBR core, not only the fuel element. The input uncertainties of the mass flow rate, inlet temperature of coolant, and the averaged power density in the fuel element should be reduced by further optimization of the PBR core.

Although the non-parametric approach is used in this paper to evaluate the tolerance limits at a given level, the parametric approach can also be applied to verify the evaluation. The three output variables are all assumed as normal distributions, which are qualified by the Kolmogorov-Smirnov (K-S) test and represented by the solid line in the Fig. 17, Fig. 18 and Fig. 19. According to the qualified normal distribution, the two-side tolerance intervals at the given (95%, 95%) level are [3018 K, 3305 K], [374.4 kPa, 443.8 kPa] and [2882 K, 3154 K] for maximum temperature, pressure drop and outlet temperature, which are relatively close to the results by the non-parametric approach. The good agreements indicate that both the non-parametric approach and the parametric approach are reliable in predicting the tolerance limits, but the non-parametric approach provides a more conservative evaluation.

5. Conclusions and future work

This work develops a procedure which combines the multi-objective optimization and uncertainty quantification to prepare an optimal fuel element for PBR with regard to the thermal performance. In the optimization process, a general shape function of the porosity profile on the cold frit is established, in which the coefficients are determined by the NSGA-2 process. The improved thermal performance of three PBR fuel elements, i.e., a bare fuel element of a symmetric power distribution, a bare fuel element of an asymmetric power distribution, and a fuel element with top and bottom reflector has verified the optimization approach. In addition, the thermal performance of the optimal design under a perturbed condition is also evaluated by the UQ. By propagating the input parameters uncertainties including the inlet temperature of the coolant, mass flow rate, averaged power density, and the pore size on the cold frit, the two-sided tolerance limits of maximum pressure, pressure drop and the outlet temperature of the bare fuel element of symmetric power distribution are derived.

However, there still needs some improvements. Presently, no nuclear feedback is included in the framework, hence it will be more appropriate and rigorous to perform the optimization considering both the thermal hydraulics and the neutronics. Besides, the objective functions are evaluated by the CFD calculation directly, which limits the population size in the NSGA-2 process due to its extensive computational effort. A surrogate model of good accuracy, therefore, should be constructed to replace the CFD calculation. Moreover, only four input parameters are selected for UQ in the post evaluation, which are just a part of the parameters affecting the thermal performance of fuel element, thus it is required to include more important parameters to reflect the actual situation in future.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.nucengdes.2019.110316.

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